

Feature Encoding Brief

Feature encoding is used to convert categorical or textual data into numerical form so that machine learning models can understand it.

Label Encoding

In label encoding, each category is given a unique number.

Example:

- Monday = 0
- Tuesday = 1
- Wednesday = 2

It creates an artificial order between values, which is not always correct for real-world data.

One-Hot Encoding

In one-hot encoding, each category is converted into a binary vector.

Example:

- Monday = [1, 0, 0]
- Tuesday = [0, 1, 0]
- Wednesday = [0, 0, 1]

This method does not create any order between categories, so it is more suitable in most cases.

Use of raw strings in ML

Machine learning models work on numbers and mathematical calculations. Raw string values like “Monday” or “January” cannot be processed by models because they are not numeric, so they must be converted into numerical form first.

Conclusion

Label encoding is simple but can create wrong assumptions of order, while one-hot encoding is better because it treats all categories equally. That’s why categorical data must always be converted before using it in machine learning models.